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Case study: a hybrid plant in Hat Yai whose good output was hidden by a wrong capacity setting

18 October 2026 · Rev04 · Prepared by Solar ESS from remote monitoring data. Customer details removed.

Summary

The plant is working well. It was registered in its monitoring app as 10 kWp, but it has a single PV string of 10 × 580 W panels, 5.8 kWp, so the app made it look as if it produced only 63% of what it should. Measured against 5.8 kWp, it yields about 4.1 kWh per kWp per day, which is good for Hat Yai. Since a battery was added on 18 May 2026, the shop has covered 98% of its own electricity.

Two items need attention first:

1. **Export was blocked for 24 days (28 Mar – 20 Apr 2026).** The inverter held production down to the shop's own use, losing about 500 kWh.
2. **The registered capacity is still not right.** It was changed from 10 kWp to 6 kWp on 18 Oct 2026; it should be 5.8 kWp, so the app's performance figures are meaningful.

Plant at a glance

Review of all monitoring data from commissioning on 29 Jan 2026 to 17 Oct 2026 (262 days), from SolisCloud.

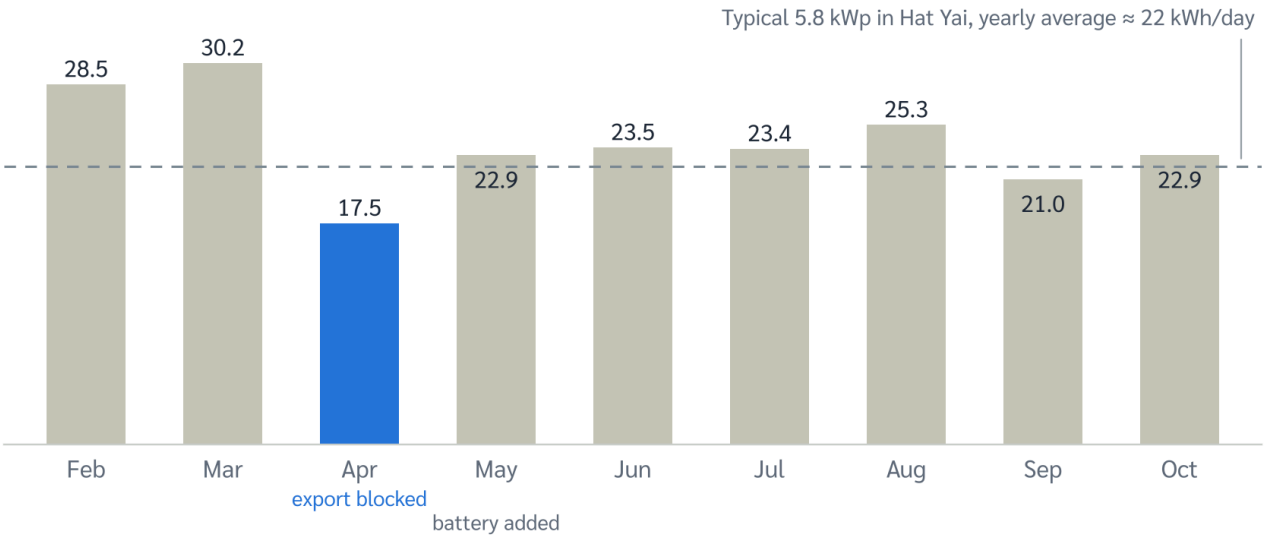
Item	Value
Location	Hat Yai, Songkhla
PV array	10 × 580 W panels in one string on MPPT1 = 5.8 kWp
Capacity in SolisCloud	10 kWp until 18 Oct 2026, now 6 kWp; should be 5.8 kWp
Inverter	Solis S6-EH3P10K hybrid, 10 kW, three-phase; MPPT2 unused
Battery	One 48 V battery, added 18 May 2026
Total production	6,230 kWh (about 4.1 kWh per kWp per day)
Best day	36 kWh (25 Mar 2026), about 6.2 kWh per kWp
Total consumption	6,183 kWh (about 22–23 kWh/day)
Grid import / export	1,402 kWh / 1,755 kWh
Self-sufficiency since battery	98% (import 0.5 kWh/day, was 12.4)
Active alarms	None

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Monthly performance

Production is close to or above a typical 5.8 kWp plant, except in April

Average 23.9 kWh/day, Feb–Oct 2026, 109% of about 22 kWh/day for a typical 5.8 kWp plant · unit: kWh per day



Source: SolisCloud daily totals, Feb–Oct 2026 (Oct = 1–17 Oct). The typical value is an approximate yearly average.

Production is close to or above a typical 5.8 kWp plant in every month but April, and well above it in the dry season. April is low because export was blocked for most of the month.

Findings

1. The registered capacity is too high

The plant has one PV string of 10 × 580 W panels (5.8 kWp), connected to input 1. At midday on sunny October days its output levels off at 4.6–5.3 kW, that is 79–91% of 5.8 kWp, normal in the heat. The best day so far, 36 kWh, is about 6.2 kWh per kWp. Divided by the 10 kWp registered until 18 Oct 2026, the same output looked like a 37% shortfall. The capacity is now set to 6 kWp; it should be 5.8 kWp.

2. Export was blocked for 24 days

From 28 Mar to 20 Apr 2026 the plant exported nothing, and each day's production equalled the shop's daytime use. Production fell to 11 kWh/day, against 30–34 kWh/day in the two weeks before and after. That is about 500 kWh lost. The inverter's operation log in SolisCloud is empty, so the setting was changed on the inverter itself or from the installer's app.

3. Selling back to the grid

The plant has exported 1,755 kWh since January. Without a PEA sell-back contract, this energy earns nothing.

4. Battery energy counters do not add up

Since May the battery shows 1,832 kWh discharged but only 1,526 kWh charged. A battery cannot give out more than it takes in, so its readings are wrong. SolisCloud lists the battery maker as "OTHER". If the inverter misreads the battery's charge level, the battery can be over-discharged or never fully charged.

5. Steady overnight load

The shop draws about 0.7 kW all night, every night: about 8–9 kWh, close to 40% of daily use. Every 0.3 kW cut saves about 2.6 kWh of battery each night.

6. Inverter runs hot

The inverter's internal temperature was 58 °C at only 2 kW output (18 Oct 2026, 11:37). Inverters reduce output when they get too hot.

7. Earnings figures in the app

SolisCloud values every kWh produced at a flat 4.7 THB and grid power at 0 THB. The earnings shown in the app are not meaningful until import and export prices are set.

8. Room to expand

The 10 kW inverter's second input (MPPT2) is free. A second string of 8 more 580 W panels (4.6 kWp) could add roughly 18 kWh/day on average. The battery already covers the night, so most of that would be exported; it pays only with a sell-back contract.

Recommended actions

Priority	Action	Expected benefit
1	Ask the installer why export was blocked from 28 Mar to 20 Apr, and lock the export setting	Prevents another loss like the 500 kWh in April
2	Set the plant capacity in SolisCloud to 5.8 kWp (now 6 kWp) and enter 10 modules × 580 W	Accurate performance figures and alerts
3	Apply for (or confirm) a PEA sell-back contract	Paid export instead of free export
4	Check the battery brand setting and the battery-to-inverter communication; recalibrate the charge level	Correct charge readings and longer battery life
5	List what runs overnight and switch off what isn't needed	About 2.6 kWh of battery saved per night for each 0.3 kW cut
6	Check the inverter's ventilation and shading from direct sun	Avoids output cuts on hot days
7	Set the import and export tariffs in SolisCloud	Correct earnings figures in the app

8	Once sell-back is in place, consider a second string of about 4.6 kWp on MPPT2	Roughly 18 kWh/day more on average

Notes on the data

- Source: SolisCloud daily totals since 29 Jan 2026, and 5-minute inverter data for 1 and 17 Oct 2026 (older 5-minute data is no longer stored).
- Array: 10 × 580 W panels, as advised by the owner. The logged string voltage (about 360 V at midday) is a little lower than usual for 10 such panels, so an open-circuit voltage check on site (about 520 V at 25 °C) is worth doing. Typical values (about 22 kWh/day for 5.8 kWp) are approximate yearly averages for southern Thailand, not a site-specific simulation.
- Rev04 replaces earlier versions: Rev03 (10 × 550 W), Rev02 (array estimated at about 6 kWp) and Rev01 (assumed two strings and reported the plant as underperforming).
- Prepared by Solar ESS from remote monitoring only; findings 4 and 6, and the string voltage, should be confirmed on site.